

Starting at 2:10 pm

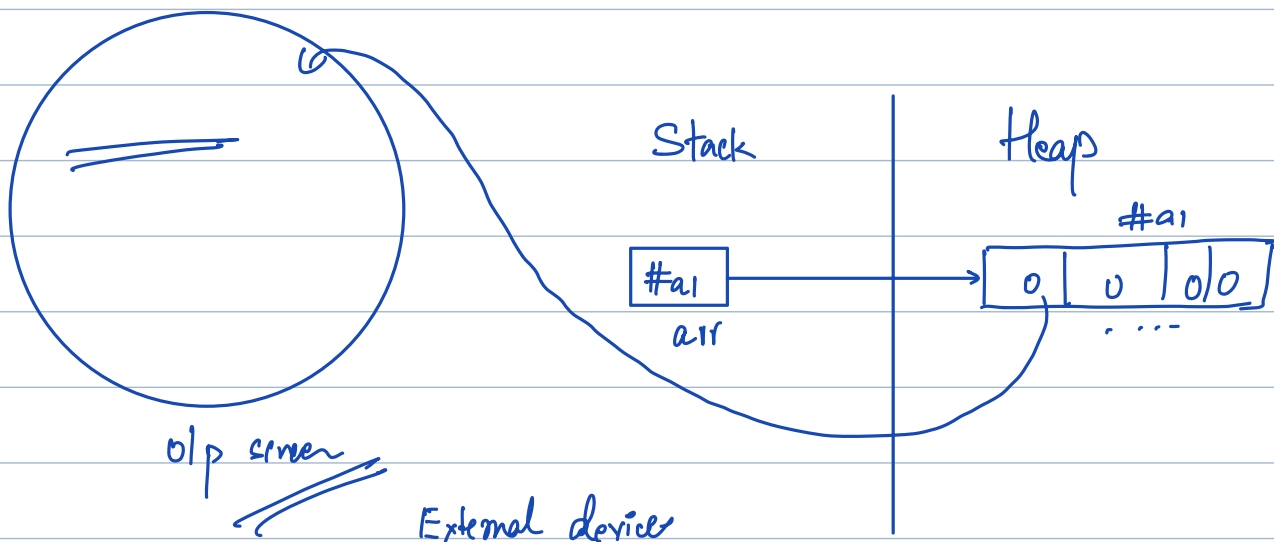
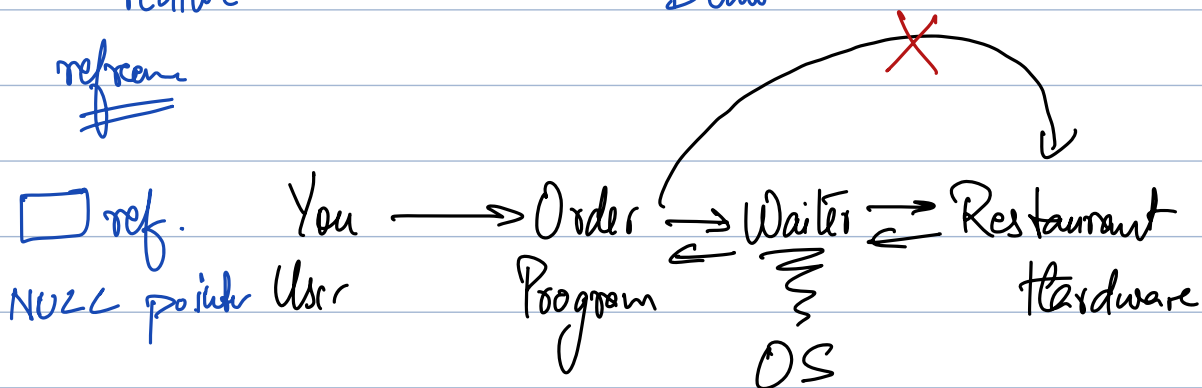
Agenda: Introduction to Operating Systems

Functions

C: calloc
malloc
realloc

System Calls

Demo



"How does this actually work inside your computer?"

```
int[] arr = new int[100];  
for (int i=0 ; i<100 ; i++) {  
    System.out.println(arr[i]);  
}
```

- Where is the array stored?
- Who gives memory for this array
- How does S.O.P point on the screen?
- How does your code even start running?

⇒ Code → simple → 1000's of operations are happening

⇒ Operating System

Ex

MacOS	}	Types of OS
Linux		
Windows		
Android		

All of them manage :

Memory	Programs	Devices
UI		

Definition

An Operating System is a manager between the user/program and the computer hardware.

```
→ int[] arr = new int[100];  
for (int i=0 ; i<100 ; i++){  
    System.out.println(arr[i]);  
}
```

1. Memory Management

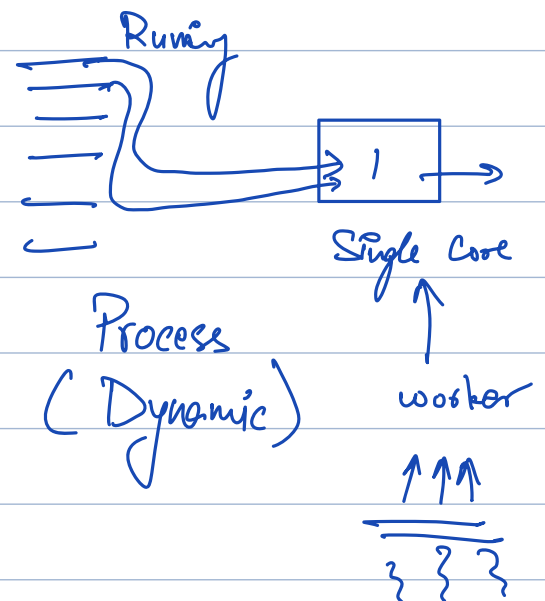
Your program (JVM) sends a request
⇒ I need space for 100 integers

OS : Is there free space in RAM?
Which part is free?
Allocates a block of memory
Marks it as "used"

Your array is stored.

2. Process Management

Program (Text) Static
→ Run → Process (Dynamic)

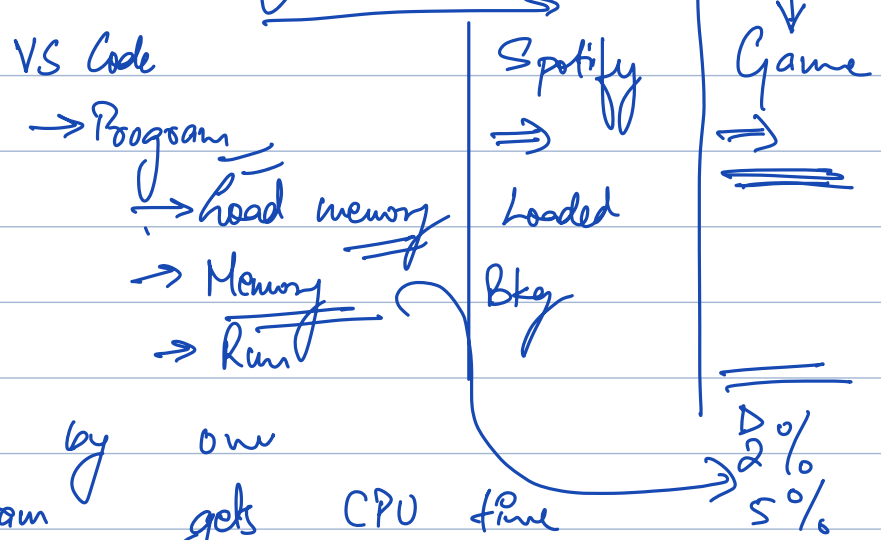


Adds it to the list of running programs
CPU Time

3. CPU Execution

```
for (int i=0 ; i<100 ; i++) {
    System.out.println(arr[i]);
}
```

CPU executes just one by one
 OS ensures → program gets CPU time
 → other program also get fair time



P₁ 10sec

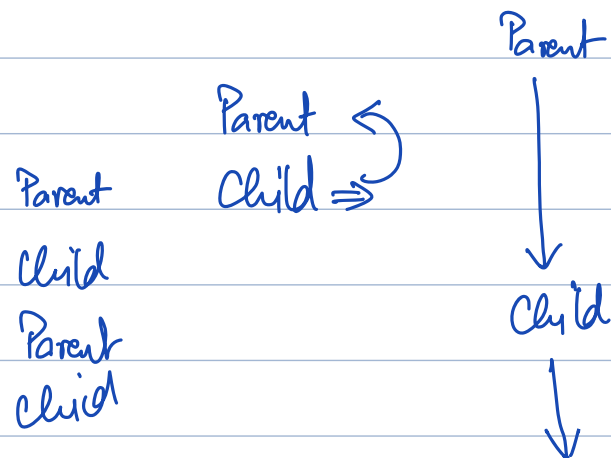
P₂ 1000 sec

P₂ 20sec

P₄ 1 sec

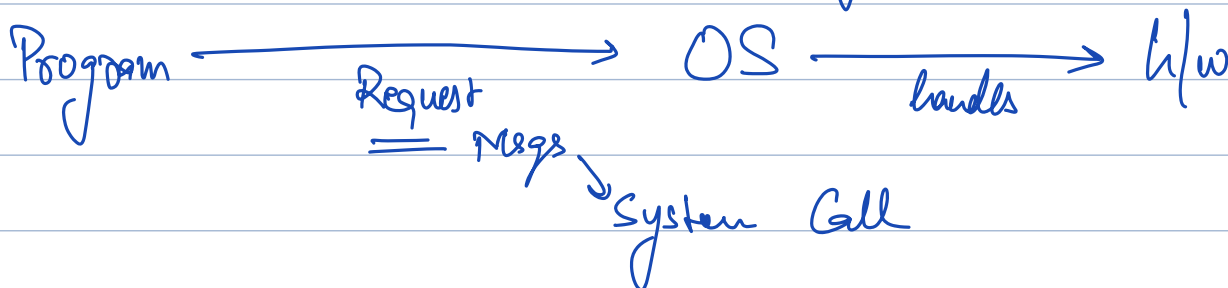
P₃

Core



4. System Calls

Your program cant directly talk to the h/w



System.out.println(arr[i]); → OS
 reqt
 ↓ sc

5. Device Management

Point something

Your program req. (System Call)

OS receive req

OS uses a device driver

Drivers communicate with the screen

Output appears

6. File Management

Save your code file

OS decides where to store the file on a disk

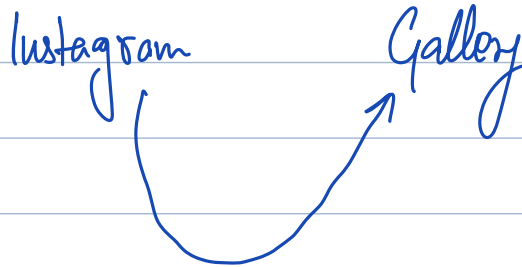
File System

Retrieve

java c
java

7. Security

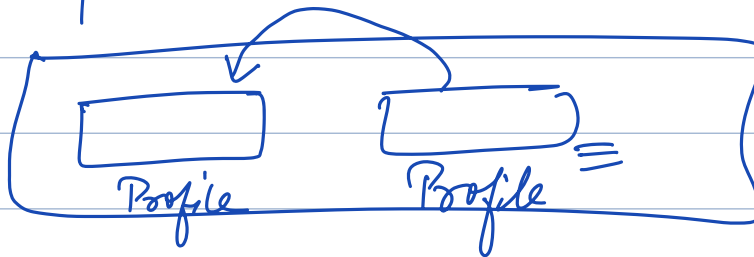
Phone



OS isolates programs

Controls permissions

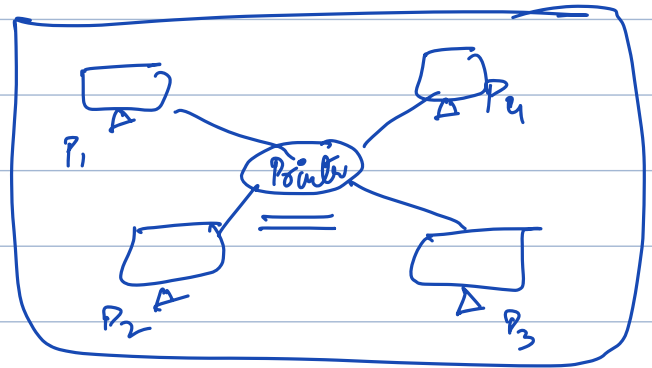
Admin



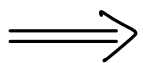
Prevent misuse

Prevent crashes

Crashes



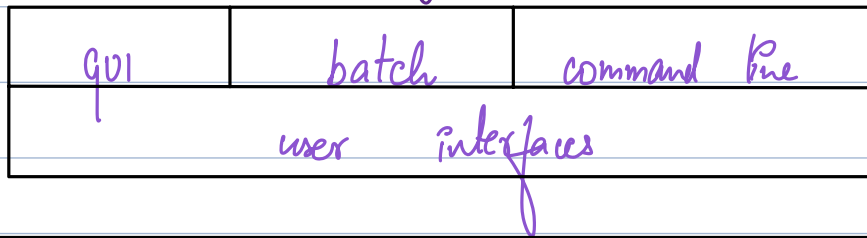
Without an Operating System, your program
cant even run



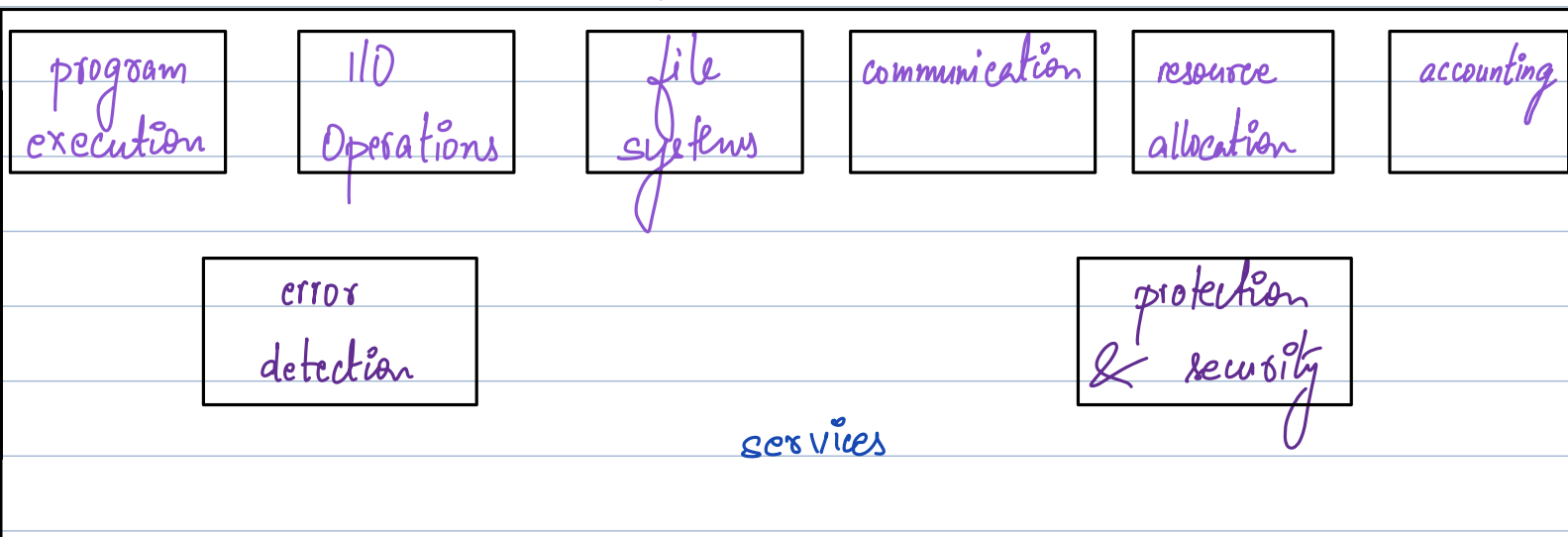
Memory
CPU time
O/p handle

H/w controlled

user and other system programs

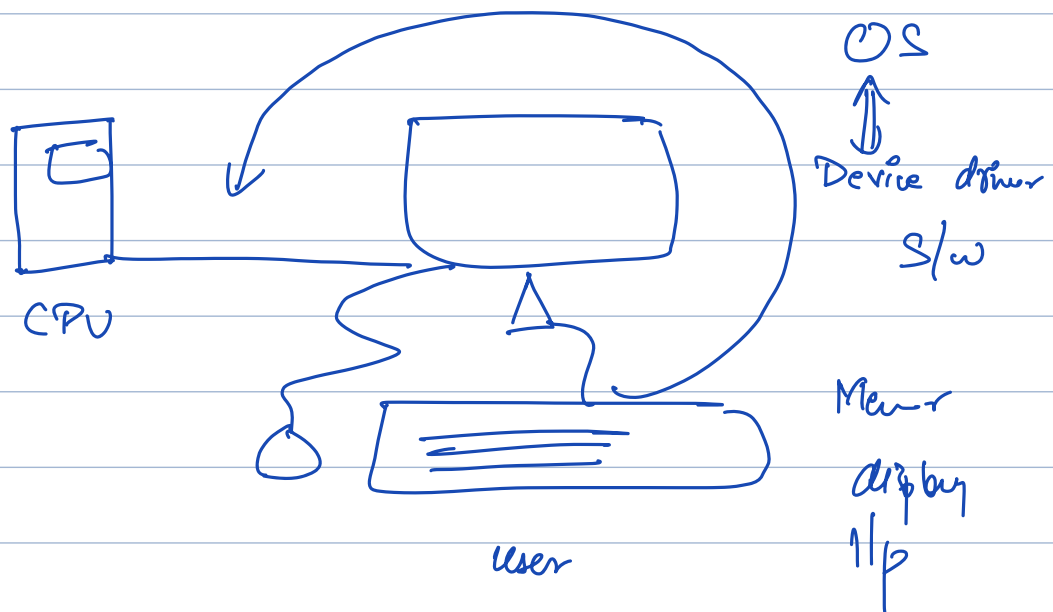


system calls



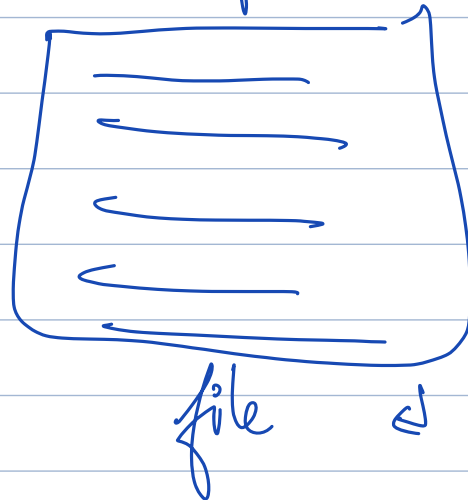
Operating System

Hardware



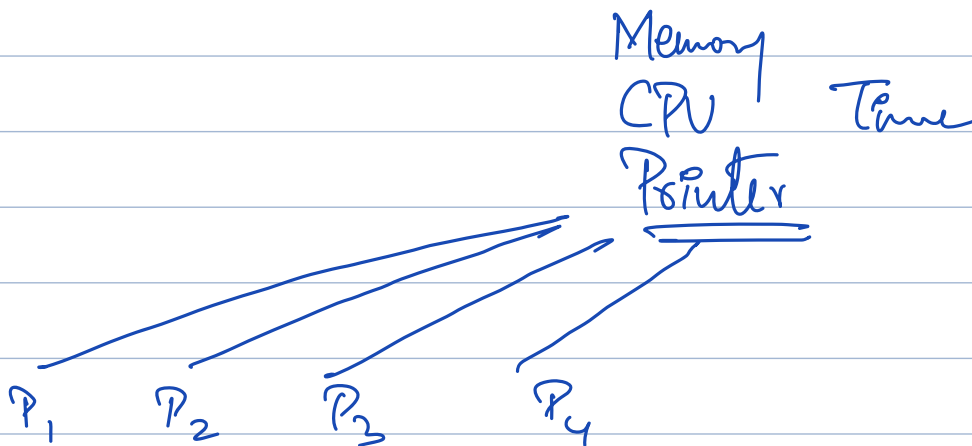
User Interfaces: GUI
CLI
TUI
Batch Interface

ls ↵
cd ↵

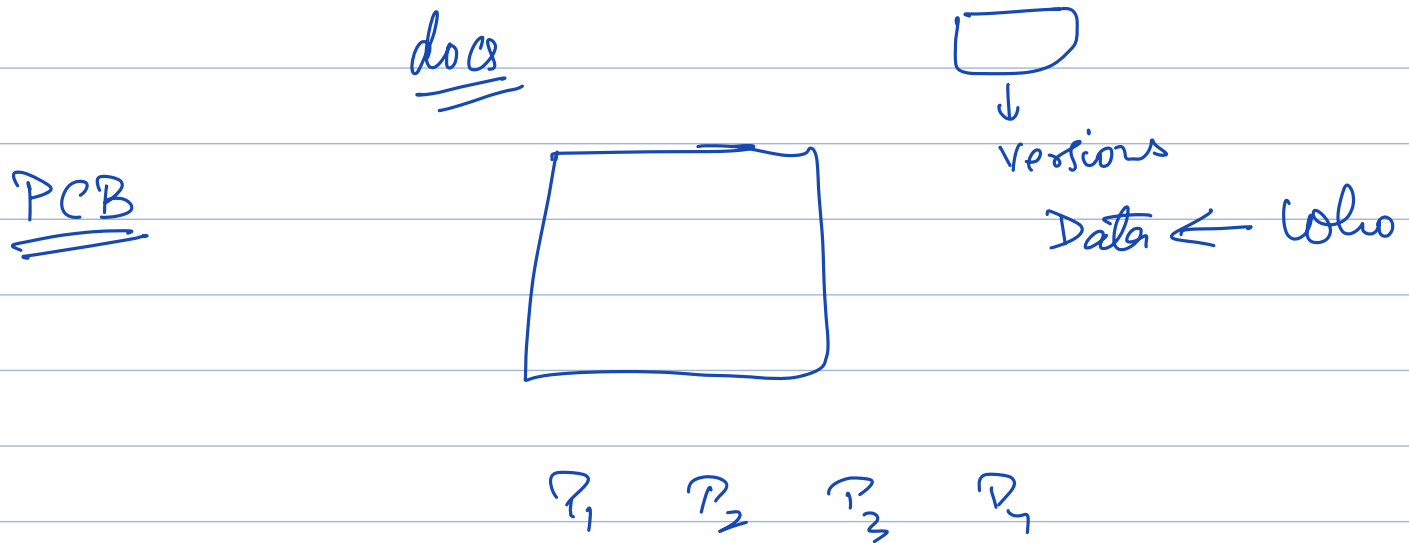
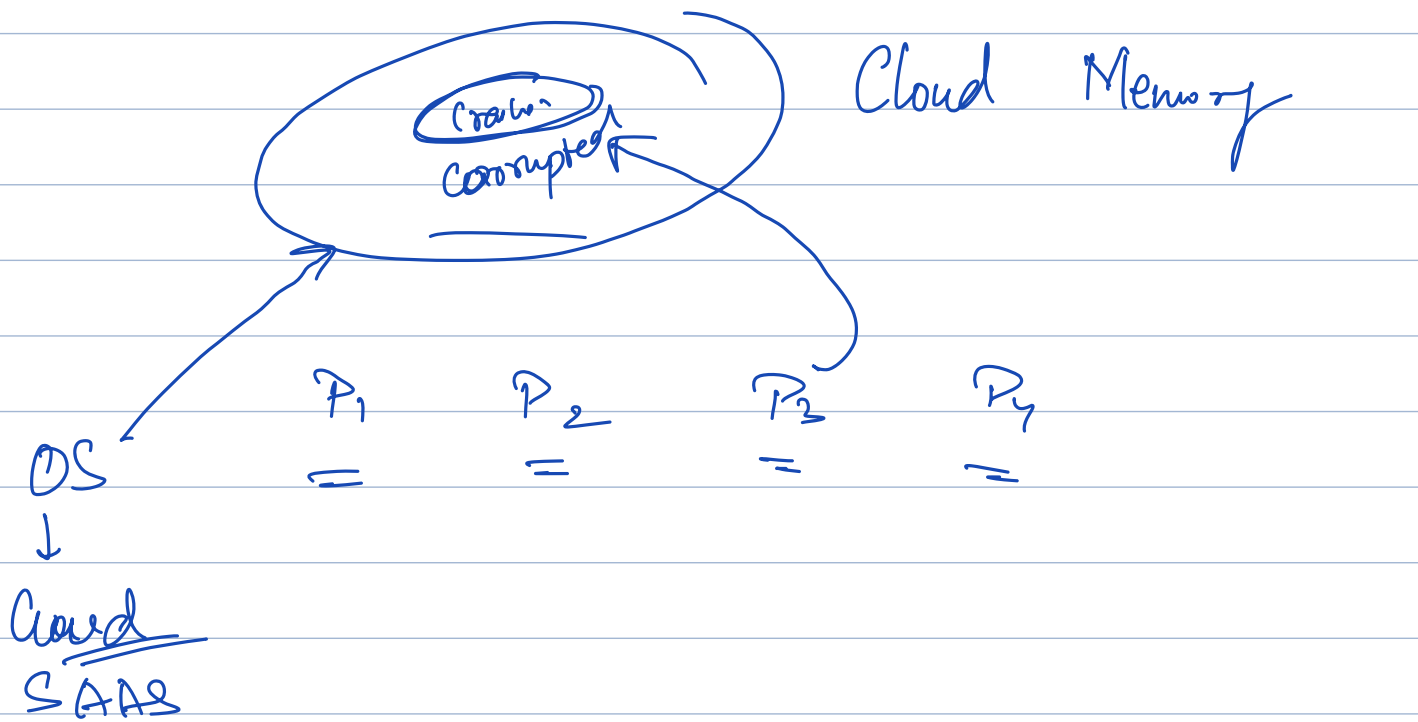


Accounting

Resource Limited



Keeps track of resource is being used
By whom ⇒ records



OS $\Rightarrow$ Interface
 Resource Manager
 Smooth & efficient execution

System Calls

System calls are the interface through which a user program requests services from the operating system.



Theory → Demo

Numericals → CPU Scheduling
+
Code → Algo
Paging

Exam

Pattern

MCQ

Numerical

Code

Project:

Shell

10%

Demo →

C

Save

Program Exam → Algo.